

Rundong Zhou

Curriculum Vitae

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RESEARCH INTERESTS

• Turbulent convection • Geophysical flows • Stability analysis • Phase transition • Numerical methods

EDUCATION

Ph.D.

University of Twente, Physics of Fluids department

Thesis: Instabilities, transitions, and turbulence in geophysical flows

Supervisors: Prof. [Detlef Lohse](#) and Prof. [Roberto Verzicco](#)

October 2024 - Present

Enschede, Netherlands

M.Sc.

Chalmers University of Technology

Joint with the University of Gothenburg, Department of Physics

Major in [Complex Adaptive Systems](#)

September 2024

Gothenburg, Sweden

Erasmus+ Exchange Program

University of Twente, Physics of Fluids department

Thesis: Swirling Kolmogorov flow

Supervisors: Dr. [Chris Howland](#) and Prof. Detlef Lohse

August 2023 - July 2024

Enschede, Netherlands

B.A.Sc. in Engineering Science

University of Toronto, Division of Engineering Science

Major in [Engineering Physics](#)

Thesis: Spectral methods in polar coordinates with discrete Hankel transforms

Supervisor: Prof. [Nicolas Grisouard](#)

June 2021

Toronto, Canada

LIST OF PUBLICATIONS

Zhou, R., Lefauve, A., Verzicco, R. and Lohse, D. *Transition to the ultimate regime of turbulent convection in stratified inclined duct flow*. Preprint, 2025. [arXiv:2508.10571](#).

Zhou, R. and Grisouard, N. *Spectral solver for Cauchy problems in polar coordinates using discrete Hankel transforms*. Preprint, 2023. [arXiv:2210.09736](#)

TEACHING

Turbulence course

University of Twente

Tutorial, assistance, and grading of assignments for students of the Applied Physics master program.

November 2025 - January 2026

Enschede, Netherlands

PEER REVIEWS

Journal of Fluid Mechanics (1), *Physical Review Fluids* (1), *Physical Review Research* (1).

CONFERENCES & TRAINING

Fall school in turbulence, Bridges over turbulent matters

Institut d'Études scientifiques de Cargèse, Université Côte d'Azur

Bridges between the various facets of contemporary research on turbulence, from mathematical fluids to experiments, from phenomenology to stochastic physics. Lectures on renormalization group, Lagrangian turbulence, quantum & wave turbulence, geophysical turbulence, etc.

October 2025

Corsica, France

2nd European Fluid Dynamics Conference

University College Dublin

In person attendance with talk given on *Transition to the ultimate regime of turbulent convection in stratified inclined duct flow*.

August 2025

Dublin, Ireland

CMiF summer school 2025

CAES du CNRS, La Vieille Perrotine

Complex motion in fluids, from micro to macroscopic. Lectures on nano fluids, sports physics, singularities, multiphase flows, planetary flows, etc.

June 2025

Oléron, France

JMBC – Multiphase flow and phase transitions course

April 2025

J.M. Burgerscentrum for Fluid Mechanics, University of Twente

Enschede, Netherlands

Lectures on melting, solidification and crystallization, immersed boundary method, phase field method, bubble dynamics, etc.

Max Planck Center Meeting

April 2025

Max Planck Institute for Dynamics and Self-Organization

Göttingen, Germany

Joint meeting program of Max Planck Center for Complex Fluid Dynamics. In person attendance with talk given on *Transition to the ultimate regime of turbulent convection in stratified inclined duct flow*.

CISM-EUROMECH Advanced School

April 2025

International Centre for Mechanical Sciences

Udine, Italy

Instabilities in turbulence. Lectures on routes to turbulence, turbulent bifurcations, nonlinear instabilities, and Rayleigh-Bénard turbulence.

Summer school in mathematics

June 2023

Université Grenoble Alpes, Institut Fourier

Grenoble, France

Topics in new trends in mathematical fluid mechanics. Lectures on mathematical analysis of incompressible fluid flow, geophysical flows, theory of water waves, singular solutions of the Euler or Navier-Stokes equations, etc.

FEATURED COURSES

Toronto: Continuum Mechanics, Computational Physics, Nonlinear Physics, Statistical Mechanics, Groups and Symmetries

Chalmers: Dynamical Systems, Non-equilibrium Processes in Physics Chemistry and Biology, Quantum Field Theory, Artificial Neural Networks

Twente: Advanced Fluid Mechanics, Turbulence, Functional Analysis, Mathematical and Numerical Physics

PROGRAMMING SKILLS

Languages: Python, Fortran, L^AT_EX, Matlab, Mathematica, C

Libraries: [Dedalus](#), [AFiD](#), [GHOST](#), NumPy, Matplotlib, SciPy